MCCJ 525 – Cybercrimes and Security

Locked or Leaked? Assessing the Security Features of Facebook, Instagram, WhatsApp, Gmail and Telegram

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Locked or Leaked? Assessing the Security Features of Five Online Platforms

1. Introduction

In Bangladesh, losing an online account is now a more common harm than losing a wallet. Around 64 million Bangladeshis use Facebook (Statista, 2025), and the Cyber Crime Awareness Foundation found that social media and online account hacking was the single most common cybercrime in the country, affecting 21.65% of the victims it studied. Roughly 59% of those victims were women, 78.78% were aged 18 to 30, and only 12% went to the police (Dhaka Tribune, 2024). Attackers are no longer breaking into computers so much as walking into accounts.

For this assignment I examined the security settings of five platforms I use every day: Facebook, Instagram, WhatsApp, Gmail and Telegram. I went through each app's settings menu on my own phone and cross-checked what I found with official help pages and recent news coverage. To compare them fairly, I rated each platform from 1 (weak) to 5 (strong) on five criteria: login protection, encryption by default, threat detection and alerts, safety of account recovery, and privacy controls.

2. Existing Security Features

2.1 Facebook

Facebook's protections sit in *Accounts Center → Password and security*. Users can turn on two-factor authentication (2FA) through an authenticator app, SMS code or physical security key, receive login alerts from unrecognised devices, and remotely log out of sessions under "Where you're logged in." A guided Security Checkup walks users through these steps. Since June 2025 Facebook has supported passkeys, which let users sign in with a fingerprint or face scan instead of a password, and in September 2025 this extended to Messenger (Meta, 2025). Profile lock, popular among Bangladeshi women, hides photos and posts from non-friends, and personal Messenger chats have been end-to-end encrypted by default since December 2023 (Crisan, 2023). The weak point is recovery: once a hacker changes the email and phone number, regaining the account is slow and support is largely automated.

Figure 1

Facebook → Accounts Center → Password and security

[Insert your own screenshot here: Facebook → Accounts Center → Password and security]

2.2 Instagram

Instagram shares Meta's Accounts Center, so it offers the same 2FA options, login alerts and login-activity list. A useful extra is *Emails from Instagram*, which shows every genuine security email Instagram sent in the last 14 days, so users can spot a phishing email pretending to be Instagram. Restrict, Hidden Words and limits on message requests help against harassment, and Teen Accounts are private by default. However, direct messages are not end-to-end encrypted. Meta withdrew even the optional encrypted chats on 8 May 2026, meaning Meta can technically access the content of DMs (Euronews, 2026). For a platform where sextortion often begins in private messages, this is a serious gap.

Figure 2

Instagram → Settings → Accounts Center → Password and security

[Insert your own screenshot here: Instagram → Settings → Accounts Center → Password and security]

2.3 WhatsApp

WhatsApp encrypts every personal chat, call and status update end-to-end by default, so not even WhatsApp can read them. On top of this sit a six-digit two-step verification PIN, passkey login, chat lock, "Silence unknown callers" and an option to hide one's IP address during calls. Since October 2025, encrypted chat backups can be protected with a passkey instead of a 64-digit key (Help Net Security, 2025). The newest addition, Strict Account Settings (January 2026), locks the account into its most restrictive state with one tap, blocking media from unknown senders and switching off link previews (WhatsApp, 2026). Its weaknesses are that the phone number is still the identity, and most WhatsApp hijacks in Bangladesh happen because users are tricked into sharing the six-digit registration code.

Figure 3

WhatsApp → Settings → Account → Two-step verification / Privacy → Advanced

[Insert your own screenshot here: WhatsApp → Settings → Account → Two-step verification / Privacy → Advanced]

2.4 Gmail

Gmail is protected by the wider Google Account. Passkeys have been the default sign-in option for personal accounts since October 2023 (Google, 2023), backed by 2-Step Verification, a Security Checkup, critical security alerts and a list of recent security activity. High-risk users can join the Advanced Protection Program, which requires a passkey or hardware key. Inside the inbox, Google states that its protections block more than 99.9% of phishing and malware attempts (Engadget, 2025), and suspicious messages carry red warning banners. Confidential mode lets users set expiry dates and block forwarding. The main limitation is that personal Gmail is encrypted only in transit, not end-to-end, so Google can scan message content.

Figure 4

Google Account → Security → How you sign in to Google

[Insert your own screenshot here: Google Account → Security → How you sign in to Google]

2.5 Telegram

Telegram offers a two-step verification cloud password, a local passcode lock, an "Active sessions" list for logging out other devices, and fine-grained privacy settings that can hide the phone number entirely and let people connect by username. Its Secret Chats are end-to-end encrypted, device-specific and support self-destruct timers. The catch is that ordinary chats, groups and channels are not end-to-end encrypted; they are stored on Telegram's servers. In September 2024 Telegram also confirmed it would share the IP addresses and phone numbers of rule-breaking users with authorities in response to valid legal requests (Engadget, 2024). Because Secret Chats must be started manually, most users never use them.

Figure 5

Telegram → Settings → Privacy and Security

[Insert your own screenshot here: Telegram → Settings → Privacy and Security]

3. Additional Features Each Platform Should Add

Every platform has the tools; the problem is that the strongest ones are optional, hidden, or written for a global audience. Table 1 lists the features I believe would most reduce the harms seen in Bangladesh.

Table 1

Proposed Additional Security Features

Platform	Proposed feature	Why it is needed
Facebook	A 48-hour "security cool-down" before a new email, phone number or 2FA change takes full effect, with alerts to the old contacts; Bangla-language human recovery support	Most takeovers succeed because hackers change recovery details within minutes; a delay gives the real owner time to react
Instagram	Restore end-to-end encrypted DMs and make them default; blur unsolicited explicit images for all users, not only teens	Sextortion and image-based abuse, which mostly target young women, start in DMs
WhatsApp	A Bangla warning screen whenever a registration code is requested; mandatory two-step PIN for new accounts	Code-sharing scams are the main cause of hijacked WhatsApp accounts
Gmail	Optional end-to-end encryption for personal accounts; lookalike-domain warnings for local banks and mobile financial services such as bKash and Nagad	Gmail is the reset key for every other account, and phishing increasingly copies local brands
Telegram	End-to-end encryption by default for one-to-one chats; a prompt to set two-step verification at sign-up; warnings on unverified investment or "job" channels	Default cloud chats expose content, and scam channels exploit users with little friction

Note. Proposals are the author's own, based on the gaps identified in Section 2.

A common thread runs through these proposals: security should be the default, not a setting that users must discover. Most people I know, myself included until this assignment, had never opened these menus, so a feature that is switched off by default protects almost no one.

4. Which Platform Is Most Secure?

Table 2

Security Scorecard of the Five Platforms (1 = weak, 5 = strong)

Criterion	Facebook	Instagram	WhatsApp	Gmail	Telegram
Login protection	4	4	4	5	3
Encryption by default	3	1	5	3	2
Threat detection & alerts	4	4	3	5	3
Recovery safety	2	3	3	4	2
Privacy controls	3	4	5	3	4
Total (out of 25)	16	16	20	20	14

Note. Scores are the author's assessment of the default and available settings as of October 2026.

WhatsApp and Gmail tie at 20 out of 25, but for an ordinary individual user I judge WhatsApp to be the most secure. Its strongest protection, end-to-end encryption, works from the moment the app is installed, with no action needed from the user. Gmail has the best account protection and phishing filters, and it deserves special care because it is the "master key": whoever controls a person's email can reset the passwords of almost everything else. Telegram scores lowest because its reputation for privacy is stronger than its defaults, while Instagram's retreat from encrypted messaging pulls it down.

Still, no platform can protect a user who hands over a one-time code or reuses a weak password. The CCAF data shows that the typical victim is young and does not report the crime, which suggests that awareness matters as much as technology. In my view, the most secure account is simply the one whose owner has switched on 2FA or a passkey, checked their recovery details and learned to recognise a scam.

5. Conclusion

All five platforms offer serious security tools, but they differ in how much of that protection is on by default. WhatsApp leads for private communication and Gmail for account protection, while Telegram and Instagram leave the most content exposed. Platforms should make strong settings the default and speak to users in their own language; users, meanwhile, should spend five minutes turning on the protections that already exist.

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